Independent branch norm transport review

Source fidelity supported; kernel unverified. Five prospective declarations were inspected against the actual imported maps and prior independent branch derivation. No compiler or fresh axiom audit was run.

Binding. Producer revision `332575b279b0`; exact candidate bytes matched the Git object. Preread plan preceded source inspection. The reviewer did not author this candidate; announced claims and earlier derivation were known.

Actual maps. The evaluator E uses the existing projective coefficient homomorphism and X coordinate. Its equality with affine polynomial evaluation followed by localization and the inverse branch-stalk equivalence is checked on constants and X. The scalar tower supplies coefficient compatibility.

Curve and norm. The quotient equation yields $E(f) = Y^2$. Mapping the polynomial identity gives

$$N^2 - E(u_1)E(u_2)E(U) = B^2Y^2,$$

and hence $(N - BY)(N + BY) = E(u_1)E(u_2)E(U)$. The factor B^2 comes from mapping $C(b^2)$ through both ring homomorphisms. The target norm equation is derived, not assumed.

Orders. At a simple branch point, nonzero b and support vanishing imply $n(a) = 0$. The predecessor gives each conjugate order one. Applying additive valuation gives

$$v_P(E(u_1)) + v_P(E(u_2)) + v_P(E(U)) = 2.$$

The source specialization derives the polynomial norm from the coefficient expressions and divisor congruences, and derives the simple-root condition from separability.

Adversarial checks. Zero b and singular branches are excluded for the order claim. A zero support polynomial would contradict the finite product order, so it cannot satisfy all premises. Characteristic two does not invalidate these local identities, but distinct-conjugate or separable-cover interpretations must not be added there. The evaluator is tied to the actual stalk; no desired norm or order is a wrapper premise.

Precise remaining work. The sum two is a stalk valuation statement. The independent hand derivation gives base-polynomial multiplicity sum one; that transport is not yet implemented. Exact compilation with complete pinned imports and independent five-name axiom readback remain required. The source file's print commands are not executed evidence. Global divisor and infinity obligations are separate.